Earth and Planetary Science Letters, 78 (1986) 296–314
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[5]

A paleomagnetic and stable isotope study of the pluton at Rio Hondo near Questa, New Mexico: implications for CRM related to hydrothermal alteration

Jonathan T. Hagstrum and Clark M. Johnson

U.S. Geological Survey, Menlo Park, CA 94025 (U.S.A.)

Received March 31, 1985; revised version received March 26, 1986

Paleomagnetic and rock magnetic data combined with stable isotope data from the middle Tertiary pluton along the Rio Hondo in northern New Mexico suggest that its magnetic remanence has both thermal (TRM) and high-temperature chemical (CRM) components. Oxygen isotope temperatures indicate that magnetite associated with the more rapidly cooled higher levels of the pluton, and with mafic inclusions and cogenetic rhyolitic dikes sampled at lower levels of exposure, ceased subsolidus recrystallization and isotopic exchange above its Curie temperature (580°C) in the presence of a magmatic fluid. Continued cooling imparted a TRM to these portions of the pluton. The more slowly cooled granodiorite at lower levels has quartz-magnetite isotopic temperatures that are below the Curie temperature of magnetite implying that its magnetization is high-temperature CRM. Sub-Curie isotopic temperatures for other granitic plutons in the western U.S.A. suggest that CRM may be commonly derived from subsolidus interactions between magnetite and magmatic fluids in plutonic rocks. A meteoric-hydrothermal system generated by the cooling Rio Hondo pluton, and not by younger adjacent intrusions, resulted in limited alteration along zones of high permeability near the southern margin of the Rio Hondo pluton, and in more prevasive alteration of the pluton to the north. The meteoric-hydrothermal alteration occurred at relatively high temperatures (> 350°C) and, with the exception of local chloritization, caused little visible alteration of the rocks. The isotopic ratios indicate that little of the magnetite could have grown from or exchanged with a meteoric-hydrothermal fluid.

1. Introduction

Chemical changes of the spontaneously magnetic minerals occur within many igneous rocks at sufficiently low temperatures to affect significantly their remanent magnetization [1]. These changes include higher-temperature oxidation and exsolution associated with initial cooling [2–4], subsequent low-temperature oxidation and maghematization [5–7], and chemically-induced volume and morphologic changes [8,9]. Chemical remanent magnetization (CRM) resulting from these changes can be stable over geologically long time periods [1,9], and is therefore of considerable importance to paleomagnetic investigations. Recognition of the magnetization process is essential to the interpretation of geomagnetic paleointensity data [4,7], and magnetization directions later modified by CRM may lead to erroneous results in conventional paleomagnetic studies.

CRM in more igneous rocks has been attributed to the post-crystallization formation of a hematite phase, which is commonly related to hydrothermal alteration. Ade-Hall et al. [10] found progressively higher Curie temperatures in titanomagnetites that were subjected to limited low-temperature (< 300°C) hydrothermal alteration, and oxidation of the titanomagnetites to titanohematite with more extreme alteration. Vincenz et al. [11] related secondary magnetic components to ilmenohematites that were formed during hydrothermal alteration of granitic rocks. Due to distinct differences in magnetic properties between cubic and rhombohedral iron-oxide phases, the relative contributions of these minerals to a rock's natural remanent magnetization (NRM) are usually decipherable with standard demagnetization procedures. More difficult to determine, however, is whether components carried by Ti-poor titanomagnetite are thermoremanent magnetization (TRM) or CRM.

Merrill and Grommé [12] found pure magnetite

in the Bucks batholith of northern California which, on the basis of thermodynamic arguments [2,13], probably formed below 600°C, and thereby acquired a high-temperature CRM. Grommé et al. [4] documented the oxidation of titanomagnetites to Ti-poor titanomagnetite in cooling basaltic lavas on Hawaii at temperatures well below their final Curie temperatures. More recently, Criss and Champion [14] compared the bulk magnetic properties of rocks from the Idaho batholith with meteoric-hydrothermal alteration patterns in these rocks as defined by stable isotope data. They suggested that reduction of magnetic susceptibilities associated with intense meteoric-hydrothermal alteration in some rocks resulted from the destruction of magnetite, even though their petrographic observations indicated that coarse magnetite was surprisingly resistant to corrosion and oxidation under such conditions. However, in less altered rocks, increased magnetic susceptibilites were thought to reflect the low-temperature growth of hydrothermal magnetite [14].

Oxygen and hydrogen isotope studies have shown that extensive circulation of meteoric-hydrothermal fluids, driven by cooling, shallowly-emplaced igneous intrusions, can cause pervasive isotopic exchange and alteration of the plutons and their surrounding country rocks [15–17]. Aside from alteration geometries and fluid origins, stable isotope ratios of rocks and minerals can also indicate water/rock ratios [15,17] and time-integrated temperature histories of fluid-rock interactions [18–20]. Because many subsolidus chemical changes of rocks occur in the presence of an aqueous fluid, stable isotope studies in conjunction with paleomagnetic and rock magnetic experiments may provide insights into the importance of CRM in igneous rocks.

This study combines paleomagnetic and rock magnetic data with stable isotope data from magnetic and silicate minerals and whole rocks in an effort to delineate the processes of magnetization during the crystallization and cooling of granitic rocks. The Rio Hondo pluton near Questa, New Mexico (Fig. 1), was selected for this study because previous work [21,22] had discovered its unusual magnetic mineralogy: the opaque minerals in the Rio Hondo pluton consist almost entirely of pure or nearly pure magnetite. A relatively pure magnetic phase was essential to this study so that meaningful isotopic measurements could be made on the magnetic separates from the rock samples. In addition, an apparent correlation between the polarity and stability of remanent magnetizations and the degree to which the Rio Hondo pluton has been hydrothermally altered suggested a possible relationship between CRM and the alteration.

2. Geologic setting

Differential uplift of the Sangre de Cristo mountains has exposed the relatively large Rio Hondo pluton to the south of the Questa caldera (Fig. 1). The Rio Hondo pluton was emplaced approximately 26 m.y. ago, and intrudes metamorphosed Early Proterozoic basement rocks which underlie the Latir volcanic field [23]. The pluton consists predominantly of granodiorite that grades into high-silica granite near its roof and toward its northern contact. Contacts at the roof generally have gentle dips, and the steep linear contact along the pluton's northeast side is localized along a major northwest-trending fault [24]. Fine-grained, rounded, mafic inclusions are widely distributed throughout the deeper levels of the pluton along Hondo Canyon. Also, the Rio Hondo pluton is cut by hundred of northwest-trending porphyritic to aphanitic dikes that range in composition from basalt to rhyolite, with rhyolite being most common. Although some dikes cut the Precambrian country rocks, they are concentrated in the Rio Hondo pluton, and probably represent intrusions of magma into already consolidated portions of the pluton during its crystallization [24].

Petrographically, the granodiorite in the deeper levels of the Rio Hondo drainage is characterized by alkali feldspar megacrysts ($\leqslant 5$ cm long) that contain inclusions of magnetite, biotite and apatite; other major phases include subhedral biotite, hornblende, sphene, and apatite. More silicic phases of the pluton lack megacrysts and hornblende. Opaque minerals constitute on average less than 2 vol.% of the granodioritic facies of the Rio Hondo pluton (B. Dillet, written communication, 1985). Magnetite is the predominant phase and occurs as fine grains (0.5–0.05 mm diameter) associated with amphibole, biotite, and sphene. In reflected-light the magnetite grains are generally

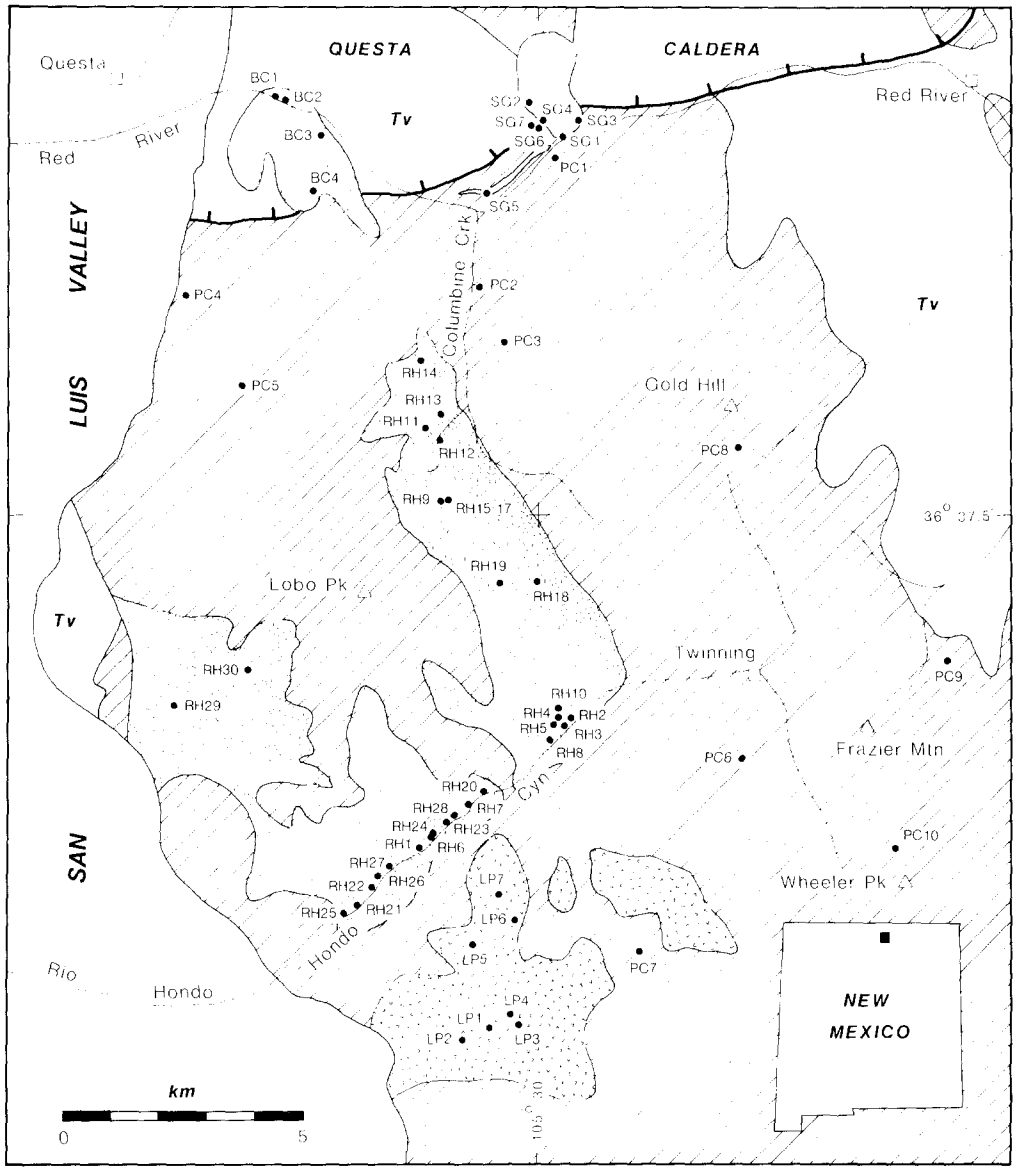


Fig. 1. Schematic geologic map of the Sangre de Cristo mountains near Questa, New Mexico, showing sample localities. Ruled areass represent pre-Tertiary rocks (mostly Precambrian), and unshaded areas east of the San Luis valley represent the Tertiary volcanic (*Tv*) rocks of the Latir field. Stippled areas indicate Tertiary intrusive rocks. Site prefixes denote rock units as follows: *RH* = Rio Hondo pluton; *SG* = Sulfur Gulch pluton; *BC* = Bear Canyon pluton; *LP* = Lucero Peak pluton; and *PC* = Precambrian basement rocks. Hachured line indicates the southern margin of the Questa caldera.

homogeneous with rare oxidation exsolution to ilmenite. Where oxidation is present, hematite is most common, but occurs only in small amounts at the rims and in minute veinlets. Ilmenite constitutes less than 5 vol.% of the opaque minerals and is usually found as rounded inclusions in sphene which generally contain minor hematite as exsolution lamellae. Hematite also occurs as minor replacement of biotite. In the granitic facies, opaque minerals usually form less than 1 vol.% of the rock. Magnetite tends to be finer grained (0.1–0.2 mm) and small crystals (< 0.01 mm) oc-

cur in feldspar. Partial oxidation of the magnetite to hematite is more developed, and ilmenite exsolution lamellae are more common than in the granodiorite.

Hydrous phases of the Rio Hondo pluton, especially hornblende, have been locally altered to chlorite. Chloritization is most pervasive in the deeper more mafic parts of the Rio Hondo pluton exposed near its southern margin in Hondo Canyon (Fig. 1). The alteration is concentrated along joints and fractures which probably formed while the pluton was still hot, as evidenced by deformed biotite and recrystallized quartz within these zones. K-Ar and fission-track ages of the Rio Hondo pluton in Hondo Canyon are discordant, ranging from 20 to 26 m.y., and may represent an emplacement age of 25-26 m.y. variably reset perhaps by nearby younger intrusions [24]. Rb-Sr mineral isochron ages of 26 m.y. determined on rocks from Hondo Canyon and the Columbine Creek drainage, support this interpretation (C. Johnson, unpublished data, 1984).

Younger plutons emplaced to the north of the Rio Hondo pluton include the Sulfur Gulch and the Bear Canyon plutons (Fig. 1). These plutons intrude the Tertiary intracaldera volcanic rocks and Precambrian wall rocks along the southern margin of the Questa caldera. The Sulfur Gulch and Bear Canyon plutons consist of high-silica porphyritic to medium-grained granite and aplite, with sparse biotite. Both plutons have been hydrothermally altered and contain local pyrite, fluorite and molybdenite mineralization which reaches economic levels in the Sulfur Gulch pluton [25]. Recalculated K-Ar biotite ages on both plutons and the mineralization cluster around 23 m.y. [26,27]. Exposed to the south of the Rio Hondo pluton is the medium-grained equigranular granite of Lucero Peak. The Lucero Peak pluton is separated from the Rio Hondo pluton by a septum of Precambrian rocks (Fig. 1), and is not cut by the extensive Rio Hondo dike swarm. The Lucero Peak pluton also contains local pyrite and molybdenite mineralization in its easternmost exposures, and concordant K-Ar and fission-track ages for the pluton are approximately 22 m.y. [24].

The northwest-trends of the Rio Hondo dikes, and of faults which also cut the Latir volcanic field to the north, suggest a middle Tertiary northeast-southwest axis of extension in the region, in contrast with the nearly east-west axis of extension associated with the present-day Rio Grande rift [28]. K-Ar ages determined on basalts intercalated with early rift-related sediments [29] and paleomagnetic data in conjunction with isotopic dates from the Latir field [21] indicate that extension along the San Luis Valley (Fig. 1) began abruptly around 26 m.y. ago. The early phase of extension was characterized in the Latir field by normal faults (now at low angles) which caused steep tilting (up to 90° NE) of volcanic units and of some resurgent plutonic rocks exposed within the caldera [21,28]. The emplacement of the Rio Hondo pluton closely followed this intense, but short-lived, early phase of extension. Paleomagnetic measurements indicate that tilting had ceased in the region by the time the other intracaldera (25–26 m.y.) and mineralized plutons (22–23 m.y.) to the north and south of the Rio Hondo pluton cooled and were magnetized [21,22]. Similar paleomagnetic directions from the Rio Hondo pluton also show no significant tilting [21] since it cooled and was magnetized, consistent with its subhorizontal roof contacts.

3. Methods

3.1. Paleomagnetic and rock magnetic techniques

Samples for paleomagnetic analysis were taken as 2.5-cm-diameter drill cores using a portable gasoline-powered drill and were oriented in the field to an estimated accuracy of 2° using magnetic and solar compasses and clinometer. Sampling was conducted to maximize the geographic distribution of sites (Fig. 1), to avoid lightning-induced magnetizations, and to cover the variation in rock type within the Rio Hondo pluton. Thus, the granodiorite, granite, mafic inclusions, and rhyolitic and andesitic dikes were all sampled. Samples were also collected from the younger mineralized Sulfur Gulch, Bear Canyon, and Lucero Peak plutons.

Cylindrical specimens 2.2 cm in length were cut from the field-drilled cores, and the remanent magnetizations of the specimens were measured with a cryogenic magnetometer. All samples were subjected to either progressive alternating-field (AF) or thermal demagnetization. AF demagnetization in peak fields between 1 and 100 mT was

the primary technique employed in this study; thermal demagnetizations were performed in air to temperatures usually in excess of 600°C. Cores 0.6 cm in diameter were drilled from representative specimens for measurement of Curie temperatures with a recording magnetic balance that determined strong-field magnetization as a function of temperature. These small cores were heated to 620°C and then cooled in steady fields of 0.2–0.5 T at rates of 10°C/minute in an argon atmosphere. To detect the presence of hematite, even that mixed with only small quantities of titanomagnetite, progressive isothermal remanent magnetizations (IRM) were measured on selected small cores using an electromagnet capable of producing a 0.7 T DC field. Polished thin sections from various sampling localities and rock types were examined in reflected and transmitted light to identify the magnetic constituents and their petrographic relations with the other mineral phases.

Components of magnetization were selected from demagnetization data using a least-squares method based on principal component analysis modified from Kirshvink [30]. Either lines (a single component) or planes (two components with overlapping coercivity or blocking temperature spectra) were fitted to the magnetization vector endpoints. The statistical methods of Fisher [31] were used to analyse the within-site line directions and unit means, and Bingham statistics [32] were used to calculate the mean direction for those sites exhibiting intersecting remagnetization circles (planes). Occasionally, one or two sample-mean directions that gave directions well away from the site-mean direction were discarded from a site-mean calculation. These samples showed either unstable demagnetization behavior or their directions were attributable to superimposed components observed in other samples from the same site that could not be resolved with demagnetization or remagnetization circle analysis.

3.2. *Stable isotopic techniques*

Isotopic ratios were measured on samples taken at the paleomagnetic sample sites within the Rio Hondo pluton, on samples from the younger mineralized plutons, and on samples collected from intervening localities in the Rio Hondo pluton and intrusive Precambrian country rocks (Fig. 1). Oxygen was extracted from the samples using ClF_3 [33], and the released oxygen was converted to CO_2 for analysis on an isotope-ratio mass spectrometer. The purity of the magnetite separates was checked by their oxygen yield during analysis. Hydrogen for mass spectrometric analysis was extracted by fusing samples at 1400°C and converting the liberated H_2O to H_2 by reaction with hot uranium metal [34]. Water contents were determined by manometric measurement of this hydrogen, and are accurate to ± 0.02 wt.% $^{18}O/^{16}O$ and D/H ratios are expressed in the standard δ notation where $\delta^{18}O$ or δD values are defined as the relative difference in the $^{18}O/^{16}O$ or D/H ratio between a sample and the SMOW (Standard Mean Ocean Water) standard expressed in parts per thousand or permil (‰). Thus a rock with a $\delta^{18}O$ value of $+6.5$ would have an $^{18}O/^{16}O$ ratio 6.5‰ greater than SMOW, and a rock with a δD value of -105 would have a D/H ratio 105.0‰ less than SMOW. All quartz and magnetite separates were analysed at least twice, and random replicate analyses of the other samples indicate that the $\delta^{18}O$ values are reproducible to ± 0.1‰. Reproducibility for the δD values is generally better than ± 2‰. The $\delta^{18}O$ value of the NBS-28 quartz standard is $+9.6$ and the δD value of the NBS-30 biotite standard is -64.5 in the Menlo Park laboratory.

4. Paleomagnetic results

4.1. *Magnetic remanence of the Rio Hondo pluton*

The Rio Hondo pluton was initially sampled for paleomagnetic study as part of a larger investigation concerning structural deformation in the Latir volcanic field related to the early development of the nearby Rio Grande rift [21,22]. Eight sites were sampled in Hondo Canyon: six in the granodiorite and two (RH3, RH8) in the northwest-trending rhyolite dikes which cut the pluton. At site RH2 in the granodiorite, samples were also taken from a relatively large mafic inclusion (1–2 m in diameter). One site (RH10, Fig. 1) was in an andesitic dike and a mafic inclusion (0.3 m in diameter) exposed in Italianos Creek. The granodiorite (RH9, RH12) and granite (RH11) were also sampled in the Columbine Creek drainage to the north.

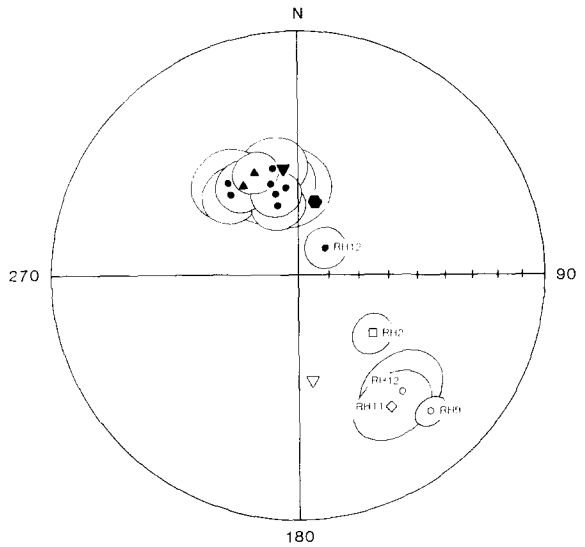


Fig. 2. Site-mean directions with 95% confidence limits for the Rio Hondo pluton. Large inverted triangle symbols indicate predicted middle Tertiary geomagnetic field directions (normal and reversed polarity) at Questa calculated from an average pole for North America [49], and hexagonal symbol indicates the present-day field direction at Questa. Small symbols denote rock type as follows: circles, granodiorite; triangles, rhyolite dikes; square, mafic inclusion (RH2); and diamond, granite (RH11). Sites RH9, RH11, and RH12, are located in the Columbine Creek drainage; the other sites are located in Hondo Canyon (Fig. 1). The samples from site RH12 have two components: a reversed-polarity component resolved with demagnetization lines and planes, and a normal-polarity component defined with demagnetization lines (see text). Diagrams are equal-area plots where unfilled symbols indicate projection from the upper hemisphere and filled symbols from the lower hemisphere.

The NRM intensities for the Rio Hondo plutonic rocks vary with rock type and range from 10^{-1} A m^{-1} for the rhyolite dikes, through 4×10^{-1} A m^{-1} for the granodiorite, to 1–2.5 A m^{-1} for the mafic inclusions and andesite dike. Usually, AF and thermal demagnetization remove a single normal-polarity component of magnetization (Fig. 2, Table 1) in the granodiorite at Hondo Canyon by 20–30 mT and 400°C (Fig. 3a, d). Median destructive fields (MDF) and blocking temperatures for this component in these rocks are low (3 mT and 150–200°C), suggesting that this magnetization is carried by coarse-grained (perhaps multidomained) magnetite. However, this same component persists to higher blocking temperatures and coercivities (550°C and 100 mT) in the finer-grained rhyolite (Fig. 3b) and andesite dikes, suggesting a primary high-temperature origin for this component carried in part by single-domain grains in these rocks. The mean demagnetization values for the dikes are also relatively low (10 mT and 200°C), indicating a significant portion of large (multidomained?) magnetite grains in these rocks as well. At site RH2, the core of the large mafic inclusion has a reversed-polarity component of magnetization (Fig. 2, Table 1) between 15 and 100 mT (Fig. 3c), although samples near its rim have a normal polarity magnetization between 20 and 100 mT and between 400 and 560°C. Samples from the mafic inclusion at site RH10 in Italianos Creek also show a reversed-polarity component (Fig. 3e), but it is too poorly defined for a site-mean calculation. Thermal demagnetizations are problematical in most rocks from the Rio Hondon pluton due to large fluctuations or steady increases in intensity upon heating above 500°C.

In general, a notable difference in magnetic remanence is observed between the samples collected in Hondo Canyon and those from the Columbine Creek drainage (Fig. 2, Table 1). All but two sites in Hondo Canyon have a single stable normal-polarity component of magnetization. The exceptions are the two mafic inclusions (RH2, RH10) which yield reversed-polarity magnetizations even though they are contained within normal-polarity granodiorite. All sites from the Columbine Creek drainage show a reversed-polarity magnetization at higher coercivities and blocking temperatures which is accompanied at site RH12 by a consistent normal-polarity component at lower coercivities and blocking temperatures. For samples from site RH9, remagnetization circles are required to define the resistant reversed-polarity component above 20 mT and 400°C (Fig. 4a, c). Demagnetization lines were fitted to those portions of the demagnetization curves representing a reversed-polarity component for samples from site RH11 (Fig. 4d). Remagnetization circles were also needed in combination with demagnetization lines (Kirschvink [30], method 2) to resolve a similar reversed-polarity component above 20 mT and 500°C (Fig. 4b, e) in samples from site RH12. However, these samples are dominated by a normal-polarity component similar to that removed from the granodiorite in Hondo Canyon (Fig. 4b).

Rock magnetic experiments on representative samples from many sites yield uniform results throughout the Rio Hondo pluton. Curie temperatures for all rock types range from 575° to 582°C (±2°C; Fig. 5), and IRM experiments indicate saturation by 0.3 T (Fig. 6). These results, in

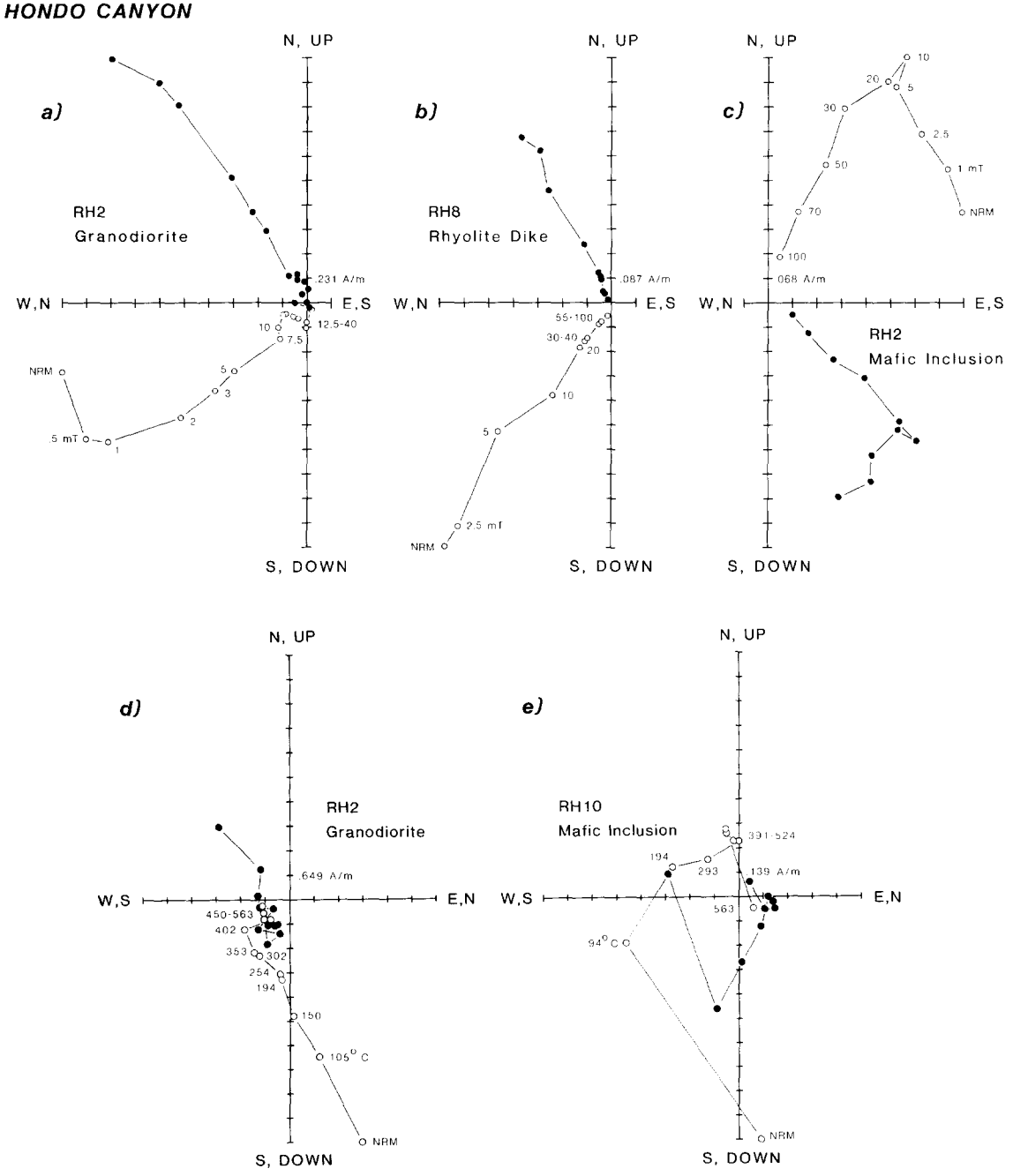


Fig. 3. Orthogonal projections of demagnetization vector endpoints for samples from the Rio Hondo pluton in Hondo Canyon. (a) AF demagnetization of a granodiorite sample from site RH2, (b) AF demagnetization of a rhyolitic dike sample from site RH8, (c) AF demagnetization of a mafic inclusion sample also from site RH2, (d) thermal demagnetization of a granodiorite sample from site RH2, and (e) thermal demagnetization of a mafic inclusion sample from site RH10. Filled symbols indicate projections of vector endpoints onto the horizontal plane, and unfilled onto the vertical plane.

TABLE 1

Site and unit mean paleomagnetic directions and poles for the Rio Hondo, Sulfur Gulch, Bear Canyon, and Lucero Peak plutons

Site	I	D	N	R	k	α_{95}	Plat	Plong
RH1	51.1	323.1	10	9.4610	17	12.2	59.3	166.5
RH2-QM	54.8	320.5	5	4.9337	60	9.9	58.2	174.1
RH2-I	−58.4	128.6	3	2.9931	319	6.9	−49.9	4.3
RH3	54.6	329.0	6	5.9067	54	9.2	65.0	170.4
RH4	57.9	343.7	8	7.5837	17	13.9	76.9	178.3
RH5	52.8	346.8	8	7.7427	27	10.8	78.7	151.9
RH6	60.3	352.4	7	6.6814	19	14.3	82.4	205.1
RH7	61.6	345.0	7	6.8905	55	8.2	76.9	197.4
RH8	52.4	337.4	6	5.9433	88	7.2	71.1	160.4
RH9	−23.8	136.1	5	4.9803	203	4.8	−43.9	324.4
RH10	65.7	343.9	12	11.5555	25	8.9	73.7	213.2
RH11	−34.5	145.1	4	3.9510	61	11.8	−54.7	324.0
RH12	77.3	43.9	14	13.6342	36	6.8	51.5	281.1
RH12-2	−36.7	138.5	10	9.6419	25	13.8	−50.5	331.9
Average	54.3	331.3	14	13.4240	23	8.6	66.7	168.7
SG1	42.2	338.7	7	6.6295	16	15.5	68.0	136.5
SG2	57.3	8.3	7	6.2841	8	22.1	83.3	331.9
SG3	51.3	353.1	6	5.7576	21	15.1	82.6	126.8
SG5	57.9	356.9	5	4.6002	10	25.4	86.9	203.0
Average	52.7	352.9	4	3.9493	59	12.0	84.0	136.1
BC1	48.8	4.2	8	7.9654	202	3.9	82.2	46.8
BC2	56.0	357.0	11	10.9246	133	4.0	87.6	161.3
Average	52.4	0.9	2	1.9946	184	18.5	86.4	64.8
LP1	56.6	53.9	10	9.9483	174	3.7	47.4	326.8
LP2	53.9	12.3	7	6.8124	32	10.8	79.8	352.4
LP3	57.4	358.7	5	4.7125	14	21.3	88.2	219.4
LP4	58.7	358.0	7	6.6672	18	14.6	72.4	180.5
LP5	50.3	6.5	5	4.5163	8	28.3	82.3	28.3
Average	57.8	9.7	5	4.8579	28	14.7	82.2	326.5

Notes: I = inclination in degrees; D = declination in degrees; N = number of samples averaged; R = vector sum; k = precision parameter; α_{95} = radius of 95% confidence in degrees; Plat, Plong = latitude and east longitude of virtual geomagnetic poles in degrees; -2 = secondary directions; Average = unit average. RH9 and RH12-2 give directions resolved with remagnetization circles and Bingham statistics [32]; for these entries, α_{95} is an average of the two Bingham 95% confidence limits, $k = k_1$ of the two Bingham concentration parameters, and R is calculated from k_1.

conjunction with the demagnetization data, petrographic work, and electron microprobe analyses (B. Dillet, unpublished data, 1985) demonstrate that the magnetic carrier in all rocks of the Rio Hondo pluton is pure or nearly pure magnetite.

4.2. *Magnetic remanance of the younger mineralized plutons*

Oriented cores for paleomagnetic analysis were collected from 5 sites in the Sulfur Gulch pluton, from 2 sites in the Bear Canyon pluton, and from 5 sites in the Lucero Peak pluton (Fig. 1). Site SG5 in the Sulfur Gulch pluton was in a large, cross-cutting aplite dike that parallels the southern caldera margin, and site SG4 was in the underground mine workings. This underground site is not considered further because its magnetization was unstable upon demagnetization. Reflected-light petrography shows that pyrite is accompanied by only minor amounts of magnetite and ilmenite in these underground rocks.

Demagnetization of samples from the other sites in these three roughly contemporaneous plu-

TABLE 2

$\delta^{18}O$ and δD whole-rock and mineral values and water contents for samples from the Rio Hondo, Sulfur Gulch, Bear Canyon and Lucero Peak plutons and the Precambrian country rocks

Site	$\delta^{18}O$	δD	H_2O^+ (wt.%)
RH1	+6.7	−91	0.37
RH2-I	+2.6		
RH2-QM	+0.8		
RH7	+0.2	−111	0.60
RH7-bio		−119	3.87
RH7-chb		−111	5.22
RH7-chh		−114	2.08
RH8-chl [a]		−123	7.06
RH11	+5.4	−103	0.45
RH12	+2.9	−110	0.55
RH13	+5.2	−103	0.27
RH14	+7.3	−97	0.21
RH15	+5.8	−118	0.66
RH16 [a]	+5.0	−108	0.78
RH17	+6.0	−117	0.40
RH18	+7.6	−98	0.22
RH19	+5.8	−117	0.48
RH20	+3.5		
RH21	+6.7	−114	0.35
RH22	+6.0	−124	0.48
RH22a	+1.3	−123	1.46
RH23	+3.5	−120	0.89
RH23a	−0.5	−119	1.09
RH24	+6.1	−125	0.59
RH25	+0.8	−118	0.71
RH26	+2.2		
RH27	+4.1		
RH28	+4.2		
RH29	+4.4	−113	0.57
RH30	+4.8	−109	0.54
SG1	+5.1	−95	0.23
SG5 [b]	+1.8	−110	0.38
SG6	+2.6	−94	0.31
SG7	+4.0	−101	0.20
BC1	+7.8	−95	0.25
BC3	+7.1	−105	0.17
BC4	+5.3	−103	0.20
LP1	+7.7	−100	0.22
LP5	+7.4	−103	0.22
LP6	+7.1	−99	0.10
LP7	+8.0	−107	0.12
PC1	+1.6		
PC2	+1.4	−108	0.82
PC3	+4.4		
PC4	+4.6	−95	0.62
PC5	+3.3	−130	0.95
PC6	+2.9	−111	1.42
PC7	+4.1	−108	0.89
PC8	+4.4	−95	2.11
PC9	+4.9	−115	1.66
PC10	+4.6	−95	0.90

tons revealed similar magnetizations (Fig. 7). Within-site dispersions are sometimes large, perhaps indicating extended cooling periods, but site-mean directions are consistent. All sites have a single normal-polarity magnetization with directions that are not significantly different (at the 95% confidence level) from the predicted middle Tertiary geomagnetic field direction. NRM intensities are approximately 10^{-2} A m^{-1} for all samples, and MDF values are around 4 mT, although coercivities are higher (MDF ≈ 10 mT) in the Bear Canyon stock and Sulfur Gulch aplite dike. Curie temperatures (579–582°C) also indicate that nearly pure magnetite is the main remanence carrier, and IRM experiments and reflected-light petrography show that it is accompanied in the Sulfur Gulch and Bear Canyon plutons by significant quantities of hematite occurring as intergrowths in the magnetite and ilmenite grains.

5. Isotopic results

5.1. ^{18}O and D variations

$^{18}O/^{16}O$ and D/H determinations on rocks and minerals provide a means for detecting and quantifying the interaction of rocks with hydrothermal fluids. Meteoric waters generally have much lower $^{18}O/^{16}O$ and D/H ratios than magmatic waters and rocks [25], and significant lowering of the isotopic ratios of rocks can result from interactions with meteorically-derived fluids. The degree to which the isotopic ratios of the rock are changed depends upon (1) the initial isotopic compositions of the rock and water, (2) the water/rock (W/R) ratio and extent to which the system has remained closed, (3) the time-integrated temperature of interaction, and (4) the mineralogy of the rock. Different minerals undergo isotopic exchange with aqueous fluids at differing rates, and the duration of meteoric-hydrothermal circulation is normally too short for equilibrium to be attained among all minerals [36,37]. Quartz and magnetite are among the most resistant minerals

Notes: I = mafic inclusion; QM = quartz monzonite; bio = biotite; chb = chloritized biotite; chh = chloritized hornblende; chl = chlorite.

[a] Rhyolite dike.

[b] Aplite dike.

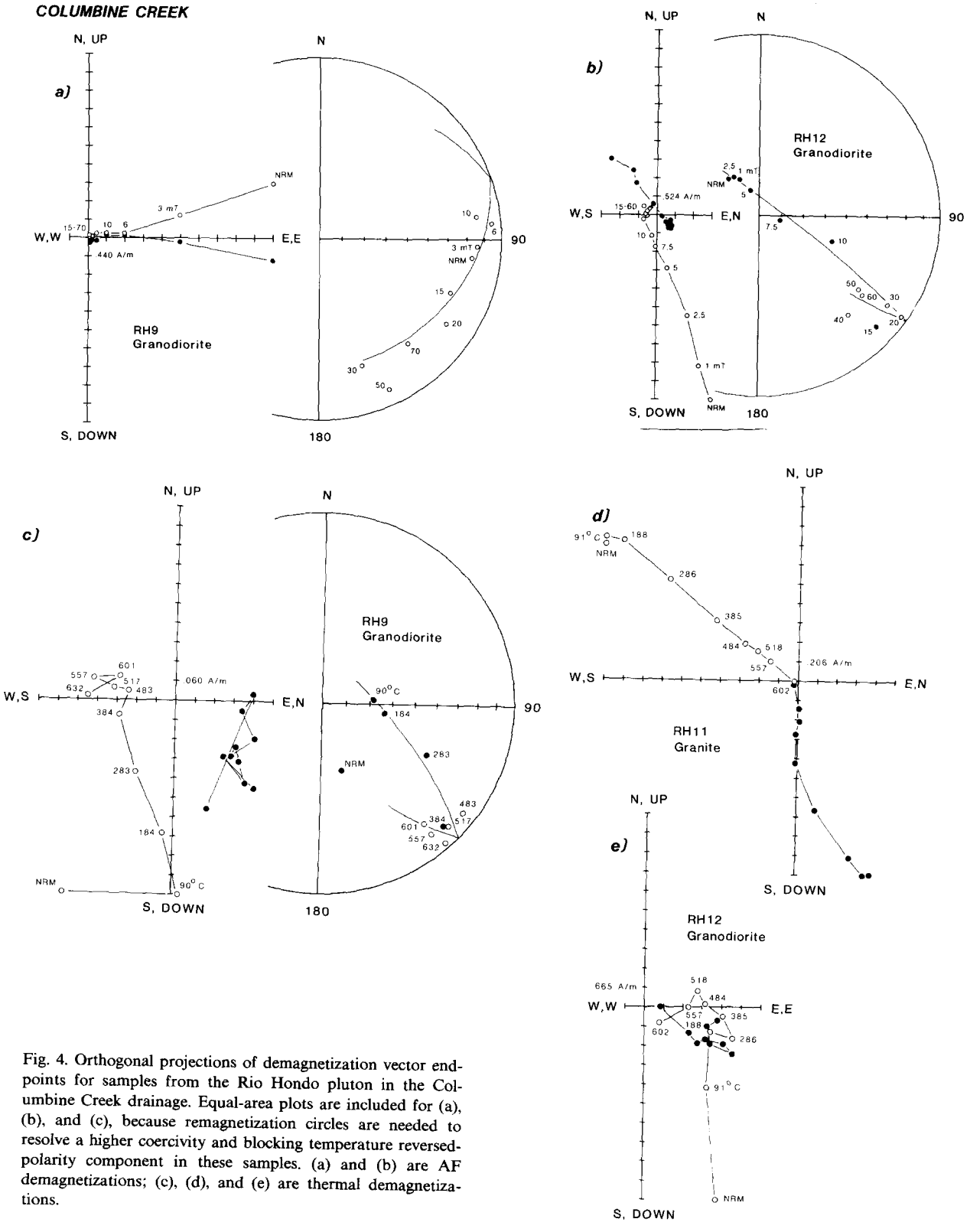


Fig. 4. Orthogonal projections of demagnetization vector endpoints for samples from the Rio Hondo pluton in the Columbine Creek drainage. Equal-area plots are included for (a), (b), and (c), because remagnetization circles are needed to resolve a higher coercivity and blocking temperature reversed-polarity component in these samples. (a) and (b) are AF demagnetizations; (c), (d), and (e) are thermal demagnetizations.

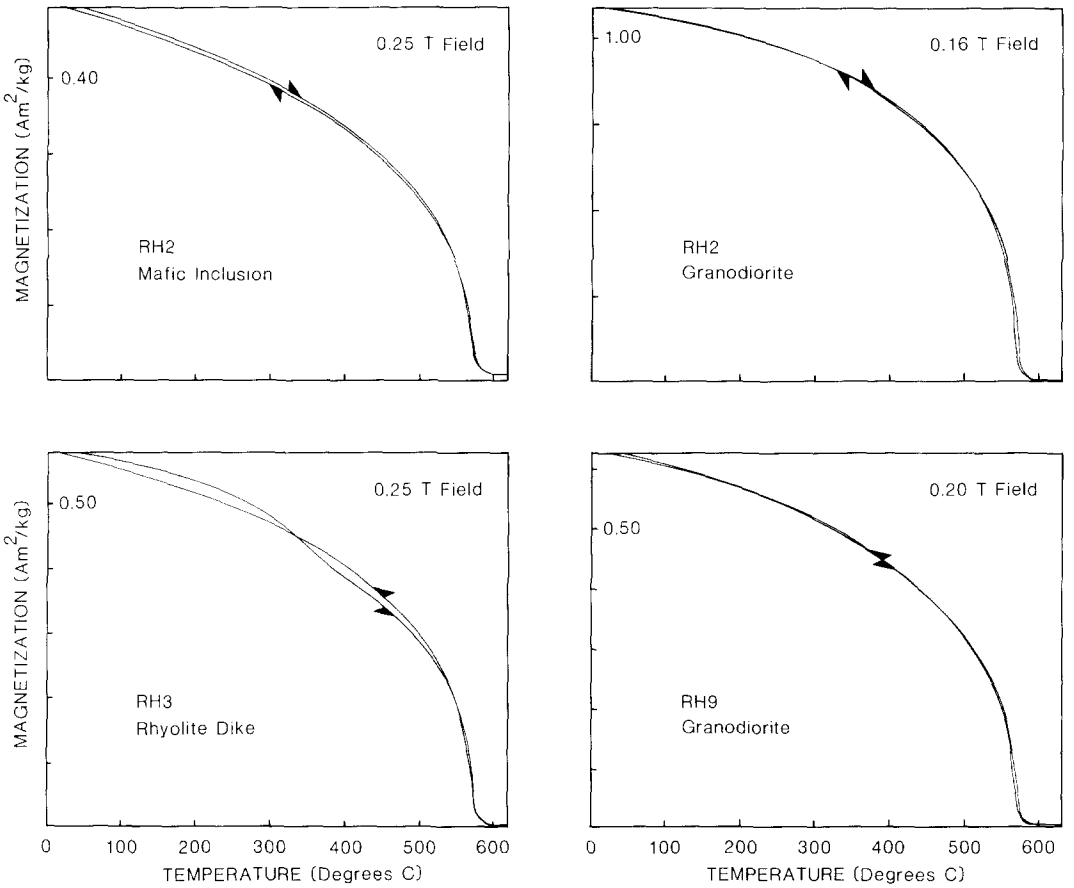


Fig. 5. Saturation magnetization vs. temperature plots for representative samples from the Rio Hondo pluton showing Curie temperatures near 580°C, indicating that the magnetic mineral in these rocks is pure or nearly pure magnetite.

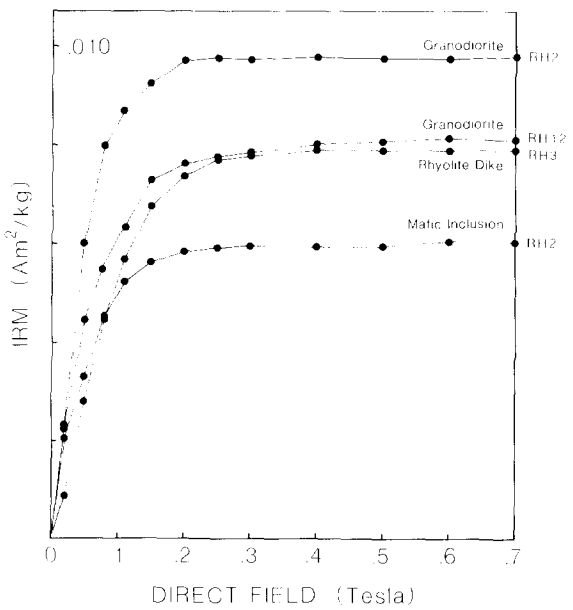


Fig. 6. Progressive IRM plots showing saturation between 0.2 and 0.3 T indicating that pure or nearly pure magnetite is the predominant magnetic mineral in representative samples from the Rio Hondo pluton.

to such oxygen isotopic exchange, whereas feldspar and biotite approach equilibrium with the fluid at a relatively rapid rate [38–40].

Whole-rock $\delta^{18}O$ and δD values (Table 2) indicate that virtually all of the Rio Hondo plutonic rocks have interacted to varying degrees with meteoric-hydrothermal fluids (Fig. 8). The interaction has shifted the isotopic composition of the rocks away from their primary magmatic values toward values characteristic of meteoric waters, dispersing the data points along a characteristic L-shaped mixing curve or band [15]. The inverted L-shape of the band results from a more rapid drop in δD to minimum values at low W/R ratios

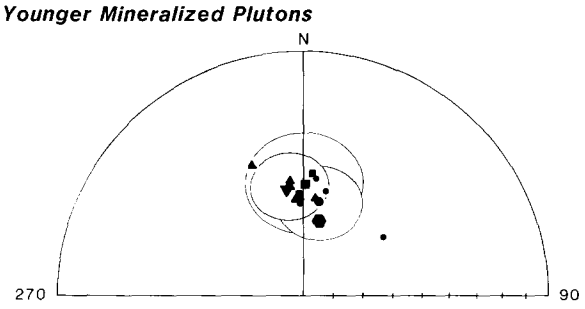


Fig. 7. Site-mean directions for the younger mineralized Sulfur Gulch (triangle symbol), Bear Canyon (square), and Lucero Peak (circle) plutons. Larger symbols denote unit means with 95% confidence limits. Inverted triangle symbol indicates the predicted geomagnetic field direction, and hexagonal symbol the present-day field direction at Questa.

because the hydrogen content of rocks is low relative to water and thus is more sensitive to the exchange process than oxygen. The atomic W/R ratios shown in Fig. 8 were calculated using the closed-system water-rock equation of Taylor [15], an assumed average temperature of 400°C, and initial δD and δ^{18}O values for the meteoric water of -90 and -12.5, respectively. The initial δD value for the water was determined at 400°C from the "equilibrium" δD rock values (horizontal portions of the L-shaped curves in Fig. 8), and with this value an initial δ^{18}O value for the water was found using the meteoric water equation of Craig [25]. The calculated W/R ratios are not strongly dependent upon temperature or closed-system/open-system behavior for values less than 0.5. Although large shifts in the δ^{18}O value for the water would be needed to appreciably affect the W/R ratios at low values, they must be considered minimum values.

Fig. 9 is an interpretive contour map of whole-rock δ^{18}O values. Some of the lowest values are associated with the Sulfur Gulch dike (SG5) and

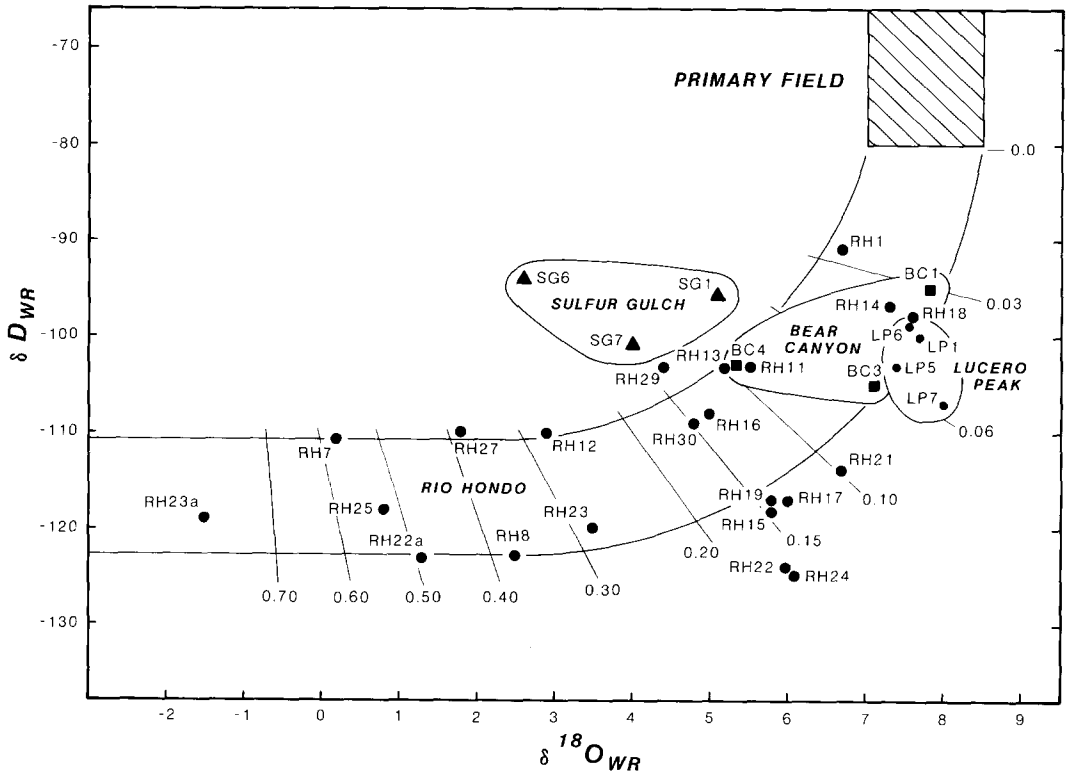


Fig. 8. Whole-rock δ^{18}O vs. δD values for the Rio Hondo (large circle symbol), Sulfur Gulch (triangle), Bear Canyon (square), and Lucero Peak (small circle) plutons. Shaded region of primary values is based on analyses from unaltered plutons associated with the Latir volcanic field, and results from other plutons throughout the world [15–17]. Water/rock ratios, increasing to the left along the mixing band, have been calculated assuming a closed system at 400°C with initial δD and δ^{18}O values of -90 and -12.5, respectively, for the meteoric water component.

Precambrian rocks surrounding the Sulfur Gulch pluton. These values are less than those within the Sulfur Gulch and Bear Canyon plutons, suggesting that the hydrothermal circulation predated their solidification or was more intense in the vicinity of the southern caldera ring fracture zone. Most of the samples from the Rio Hondo pluton in the Columbine Creek drainage have W/R ratios similar to samples from the Sulfur Gulch pluton and thus could be considered to have experienced approximately the same meteoric-water flux (Fig. 8). However, the Sulfur Gulch pluton could have been altered at higher temperatures, or perhaps with a relatively D-rich water compared to the Columbine Creek portion of the Rio Hondo pluton. The Precambrian intrusive rocks adjacent to the Lucero Peak pluton (PC7, Fig. 9) have lower $\delta^{18}O$ values than those in the pluton itself, suggesting that the pluton's hydrothermal system was driven by the intrusion before solidification. Neither the $\delta^{18}O$ value for site PC7 nor those of intrusive Precambrian rocks near the Bear Canyon pluton (PC4-5) are as low as the values near the Sulfur Gulch pluton implying that these two plutons generated smaller external meteoric-hydrothermal systems. Calculated W/R ratios for the Lucero Peak and Bear Canyon plutonic rocks (Fig. 8) are about half that of the Sulfur Gulch pluton and the Rio Hondo pluton in Columbine Creek indicating a much lower water flux through the Lucero Peak and Bear Canyon plutons. The $\delta^{18}O$ values of the easternmost intrusive Precambrian rocks are several permil lower than those of most high-grade metamorphic rocks, and are probably the result of interactions with the meteoric-hydrothermal system associated with the Rio Hondo pluton.

Isotopic measurements on closely-spaced samples from the Rio Hondo pluton in Hondo Canyon indicate that meteoric-hydrothermal alteration was highly variable (Fig. 9). $\delta^{18}O$ values range from $+6.7$ to -0.5 (Table 2), and rocks differing by more than 4‰ in $\delta^{18}O$ can be found within a few meters of each other, with the lowest values occurring in highly-fractured, strongly-chloritized zones near joints and faults. W/R rations for these permeable zones indicate that the meteoric-water flux was as much as 5 times that which passed through the Columbine Creek portion of the pluton (Fig. 8). However, rocks characterized by nearly primary magmatic $\delta^{18}O$ values have also been chloritized, suggesting two episodes of chloritization within the Rio Hondo pluton. The first generation most likely formed in equilibrium with a magmatic fluid during deuteric alteration of the pluton (this would not lower $\delta^{18}O$ values), and the second in partial equilibrium with the subsequent meteoric-hydrothermal fluids that circulated predominantly through the permeable zones of the pluton. The biotite, chloritized biotite, hornblende, and plagioclase from the most extensively meteorically-altered sample of granodiorite in Hondo Canyon (RH7) define a Rb-Sr isochron age of 25.7 ± 0.2 m.y., indicating that the meteoric-hydrothermal system in Hondo Canyon was assocated with solidification of the Rio Hondo pluton. An identical Rb-Sr mineral-isochron age (C. Johnson, unpublished data, 1984) using chloritized biotite from a sample site in the Rio Hondo pluton in Columbine Creek (RH12) indicates that meteoric-hydrothermal alteration was associated with initial cooling in that portion of the pluton as well.

5.2. *Oxygen isotope thermometry*

The partitioning of oxygen isotopes among coexisting phases depends upon temperature, and thus equilibrium fractionation temperatures can be calculated for coexisting mineral pairs from their $\delta^{18}O$ values and empirically determined fractionation factors. The constants used in this study were compiled by Bottinga and Javoy [18,19] and Javoy [20], and the $\delta^{18}O$ values and "isotopic" temperatures are presented in Table 3. The precision of these temperatures is generally $\pm 20°C$, due principally to the reproducibility of the $\delta^{18}O$ measurement, although actual errors (accuracy) may be as much as twice the precision due to the various experimental calibrations. Most quartz-magnetite temperatures are between 520° and 650°C within the Rio Hondo pluton, suggesting that significant subsolidus isotopic exchange occurred in the presence of a *magmatic* fluid. These temperatures are similar to those found in granitic plutons that did not suffer subsequent meteoric-hydrothermal alteration ([41], J.R. O'Neil, unpublished data, 1984) implying that the quartz and magnetite in the Rio Hondo pluton were resistant to oxygen isotopic exchange with the meteoric-hy-

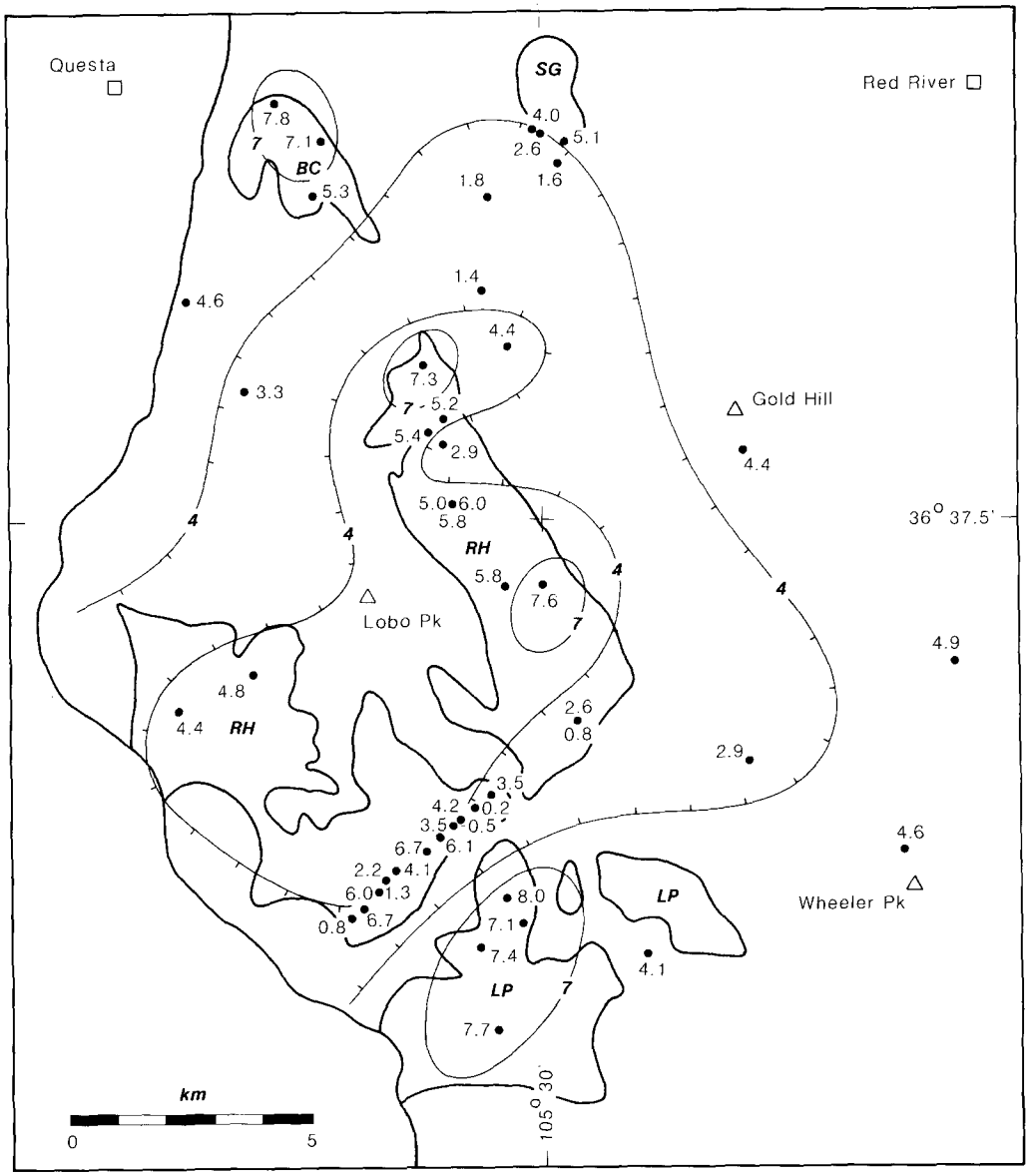


Fig. 9. Interpretive contour map of whole-rock $\delta^{18}O$ values of the Rio Hondo (*RH*), Sulfur Gulch (*SG*), Bear Canyon (*BC*), and Lucero Peak (*LP*) plutons and the Precambrian basement rocks. A zone of intense meteoric-hydrothermal alteration ($\delta^{18}O < 4$; hachured contour) encompass the Rio Hondo pluton. Relatively unaltered regions ($\delta^{18}O > 7$) coincide with the Lucero Peak and Bear Canyon plutons and with the fine-grained granitic rocks of the Rio Hondo pluton.

drothermal fluid. Lower isotopic temperatures (< 600°C) generally correspond to lower levels within the Rio Hondo pluton and probably reflect slower cooling rates. Higher isotopic temperatures (> 600°C) are found toward the more rapidly cooled roof of the pluton (RH9, RH18, RH19) and also at lower levels in the finer-grained mafic inclusion (RH2-I) and rhyolitic dike (RH8).

To assess whether or not the Rio Hondo plutonic rocks attained isotopic equilibrium, the $\delta^{18}O$ values of mineral separates (Table 3) have been plotted using the isotherm method developed by Javoy et al. [42]. These plots (Fig. 10) are constructed so that the ^{18}O fractionations of an

TABLE 3

$\delta^{18}O$ mineral values and quartz-magnetite temperatures for samples from the Rio Hondo, Sulfur Gulch, Bear Canyon and Lucero Peak plutons

Site	$\delta^{18}O_q$	$\delta^{18}O_{ksp}$	$\delta^{18}O_{mt}$	$\delta^{18}O_{bio}$	$\Delta_{q\text{-}ksp}$	$T_{q\text{-}mt}$(°C)
RH1	+8.4	+7.0	+0.6	+2.7	1.4	570
RH2-I			+1.1			600 [a]
RH2-QM			−0.6			530 [a]
RH4			+0.3			560 [a]
RH7	+5.4 [b]	−1.2	−1.3		9.6 [a]	490 [a]
RH8 (dike)		+2.5	+1.9	−5.7 [c]	5.9 [a]	650 [a]
RH9			+1.3			610 [a]
RH11	+8.6	+4.1	+0.1		4.5	540
RH12	+8.4	0.7	+0.2	−4.4	7.7	550
RH13	+8.4	+4.2	−0.4	+0.3	4.2	520
RH18	+8.4	+6.1	+1.3	+4.1	2.3	610
RH19	+8.5	+4.7	+2.0	+2.0	3.8	650
RH30	+8.4	+3.6		+1.2	4.8	
SG1	+8.3	+2.7			5.6	
SG5 (dike)	+8.2	−0.6			8.8	
BC1	+8.9	+5.8			3.1	
LP1	+9.9	+6.1			3.8	

Notes: q = quartz; mt = magnetite; ksp = potassium feldspar; bio = biotite; $\Delta_{q\text{-}ksp}$ = the permil fractionation between quartz and feldspar, approximated by $\delta^{18}O_q - \delta^{18}O_{ksp}$; $T_{q\text{-}mt}$ = quartz-magnetite oxygen fractionation temperature (see text); I = mafic inclusion; QM = quartz monzonite.

[a] Quartz $\delta^{18}O$ value assumed equal to +8.4.

[b] Recrystallized quartz.

[c] Completely chloritized biotite.

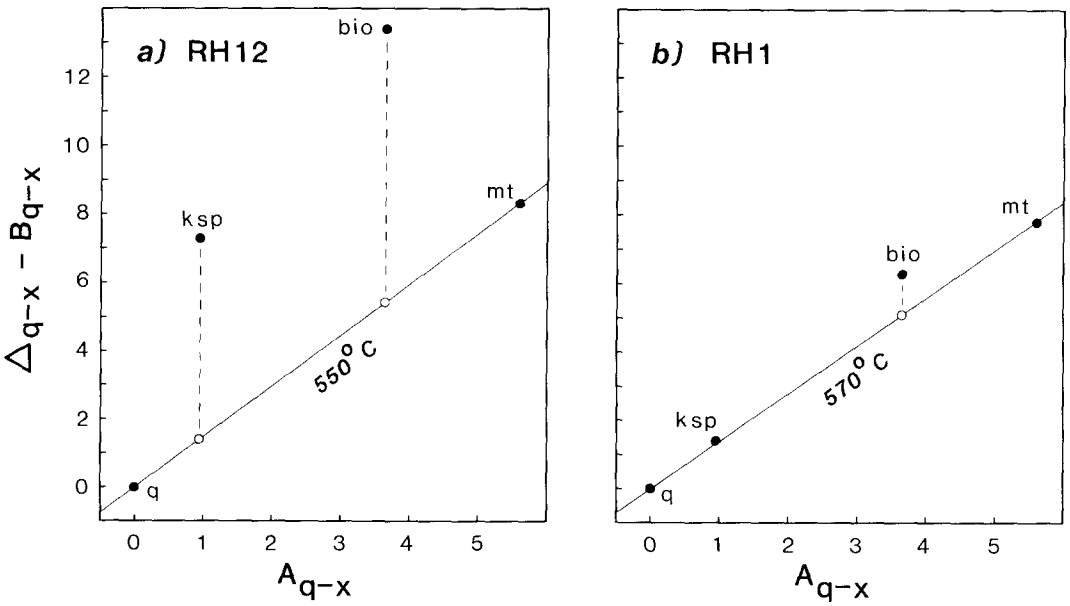


Fig. 10. Representative isotherm plots of fractionation equilibrium relationships in (a) altered (RH12) and (b) unaltered (RH1) rocks from the Rio Hondo pluton. q = quartz; ksp = potassium feldspar; bio = biotite; mt = magnetite. The plots are based on a simple re-arrangement of the basic fractionation equation $\Delta_{q\text{-}x} = A_{q\text{-}x} \times 10^6/T^2 + B_{q\text{-}x}$, where $\Delta = 1000 \ln \alpha_{q\text{-}x}$, A and B are fractionation constants (see text), and $10^6/T^2$ is the slope of the line through the points of an equilibrium assemblage. In (a), the biotite has been lowered (driven up in the isotherm plot) 8‰ from what it should be at 550°C, and the potassium feldspar has been lowered 6‰ from what it should be at 550°C. In (b), the potassium feldspar has remained relatively unaffected, but the biotite still has been lowered slightly (1‰) from what it should be at 570°C.

equilibrium mineral assemblage define a line that passes though the origin and has a slope inversely proportional to the equilibrium temperature. The plots are referenced to quartz, because it is least affected by meteoric-hydrothermal exchange. Two representative plots for rocks that are relatively altered and unaltered by a meteoric-hydrothermal fluid are depicted in Fig. 10a and b, respectively. The $\delta^{18}O$ values of feldspar and biotite are drastically lowered in the altered rocks (raising their values in the plots referenced to quartz), and the lower apparent temperatures associated with the lowest $\delta^{18}O_{qtz}$, $\delta^{18}O_{mt}$, and whole-rock $\delta^{18}O$ values suggest that quartz and magnetite may also have undergone limited isotopic exchange with meteoric water in the most extreme cases (e.g. RH7). However, the $\delta^{18}O$ values of magnetite indicate that, in general, no more than 5–12% of this mineral actually grew from or equilibrated with a meteoric-hydrothermal fluid.

6. Discussion

The Rio Hondo pluton has undergone two stages of hydrothermal alteration during its solidification and cooling. The first stage involved high-temperature recrystallization and isotopic exchange with a magmatic-hydrothermal fluid. During this early recrystallization, titanium was probably moved from the original titanomagnetite lattice to sphene, which commonly forms irregular coronas around magnetite in the Rio Hondo pluton. Zones near the roof of the pluton, the lower-level mafic inclusions, and the cross-cutting rhyolite dikes apparently cooled more quickly and quenched this subsolidus recrystallization above the Curie temperature of the remanence carrier (magnetite, 580°C). Slower cooling rates at deeper levels within the granodiorite prolonged the recrystallization and isotopic exchange to temperatures between 520° and 570°C (Table 3).

The second stage of alteration involved meteoric-hydrothermal fluids. Fracturing and faulting of the pluton in Hondo Canyon allowed meteoric waters to penetrate these high-permeability zones. Isotopic exchange and recrystallization associated with this circulation in general did not significantly effect quartz or magnetite, although it was intense enough at site RH7 to lower the $\delta^{18}O$ values of quartz from +8.4 to +5.4 and of magnetite from +0.6 to −1.3, assuming that site RH1 represents the unaltered equivalent. The effect of meteoric-hydrothermal alteration on the Rio Hondo pluton in the Columbine Creek drainage was more uniform (Figs. 8, 9). The unusual variability of the meteoric-hydrothermal alteration in Hondo Canyon and the concordance of alteration mineral ages with the crystallization age of the Rio Hondo pluton in Hondo Canyon and within the Columbine Creek drainage, preclude the imposition of this hydrothermal system by the younger adjacent Lucero Peak, Sulfur Gulch, or Bear Canyon plutons. However, the Hondo Canyon rocks could have been affected by conductive reheating or magmatic fluids related to intrusion of the nearby Lucero Peak pluton.

The quartz-magnetite isotopic temperatures for the Rio Hondo pluton suggest that the NRM of the higher-level zones, the larger mafic inclusions, and the rhyolite dikes is TRM, and that the NRM of the lower-level granodiorite is at least in part high-temperature CRM. The paleomagnetic data exhibit a similar trend: the higher-temperature rocks (e.g. RH2-I, RH9) tend to contain a higher coercivity and blocking temperature reversed-polarity magnetization and the lower-temperature rocks (e.g. RH2-QM, RH4, RH12) tend to be dominated by a less resistant normal-polarity magnetization. Thus, it appears that the higher-temperature portions of the pluton were initially magnetized in a reversed geomagnetic field. As the pluton continued to cool, the field switched polarity and the lower-temperature portions of the pluton were subsequently magnetized. The cross-cutting rhyolite dikes, which have an isotopic temperature of 650°C and a normal-polarity magnetization (RH8), were apparently emplaced after this switch in polarity.

Recent microanalysis of the NRM carried by individual oriented mineral grains from the Tatoosh granodiorite [43] and previous work on mineral separates from other intrusive rocks [44–46] have shown that much of the stable remanence in these rocks is carried by very fine-grained magnetite included in silicate minerals, usually plagioclase feldspar. Wu et al. [43] also demonstrated that the remanent magnetization associated with coarse-grained magnetite in ferromagnesian minerals from the Tatoosh intrusion was generally unstable. Dodson et al. [47] found no change

in the saturation magnetization (J_s) of a feldspar separate from the Tatoosh intrusion upon heating to 600°C, whereas whole-rock samples showed significant increases in J_s upon heating to the same temperature under the same conditions. From these results they concluded that the magnetite in feldspar was initially in equilibrium with the rock to 600°C and probably carries a TRM, and that the larger-grained magnetite associated with ferromagnesian minerals is of lower-temperture origin.

In light of the above work on the Tatoosh intrusion and other intrusive rocks, the isotopic measurements and temperature calculations for the Rio Hondo pluton could be biased by the larger volume of coarser-grained magnetite in a bulk magnetite mineral separate and therefore may not be representative of the smaller portion of fine-grained magnetite that carries most of the stable remanent magnetization. This situation could be the case for site RH12 and RH11. The NRM at site RH12 consists of a normal-polarity component superimposed on a more resistant reversed-polarity component (Fig. 4b, e). The isotopic temperature for this site suggests that the bulk of the magnetite in these rocks equilibrated with a magmatic fluid around 550°C (Table 3) and is probably associated with the dominant normal-polarity component. The reversed-polarity magnetization in samples from site RH12 could reside in an amount of magnetite too small to influence the isotopic measurements and have instead a higher-temperature origin. For site RH11, a reversed-polarity magnetization suggesting higher-temperature origins (Fig. 4d) is also found in conjunction with a low isotopic temperature (540°C). Three samples with an intermediate direction were discarded from this site implying that either these rocks were magnetized during a polarity transition or, more likely, that they carry two (normal and reversed) superimposed components. Again, the isotopic ratios of the bulk magnetite separate may not represent those of the portion which carries the reversed-polarity magnetization. However, for those samples with an isotopic temperature above 580°C, a lower-temperature alteration of only the finer-grained magnetite is considered unlikely. O'Neil and Taylor [48] have found that cation exchange greatly facilitates oxygen isotope exchange in silicate minerals. Thus, coarse-grained magnetite associated with ferromagnesian minerals would be more susceptible to isotopic exchange than fine-grained magnetite in feldspar, and any significant alteration of magnetite would be noticeable in the isotopic measurements.

7. Conclusions

Interactions between cooling plutonic rocks and hydrothermal fluids may lead to remanent magnetizations that include both TRM and CRM. Sub-Curie isotopic temperatures (< 580°C) from other granitic plutons in the western U.S.A. ([41], J.R. O'Neil, unpublished data, 1984) suggest that CRM may be commonly derived from interactions between magnetite and magmatic fluids. These higher-temperature interactions are not easily recognized because they do not necessarily include mineralization or even simple alteration. Contrary to the inferences drawn by Criss and Champion [14], we find no evidence for significant growth or recrystallization of magnetite during meteoric-hydrothermal events. This work suggests that paleointensity studies are dubious at best in slowly-cooled plutonic rocks, and could yield erroneous results if the rocks are considered to carry only TRM.

Acknowledgements

We thank J.R. O'Neil for the generous use of the USGS Stable Isotope Laboratory in Menlo Park. The authors were introduced to the geology of the Latir volcanic field by P.W. Lipman. J.C. Reed, Jr. provided many of the Precambrian samples used for isotopic analysis, and several others in the Tertiary plutonic rocks were collected with the assistance of G.K. Czamanske. D.P. Elston, P.W. Lipman, and G. McGimsey assisted in collecting the paleomagnetic samples, and B. Dillet shared the results of her reflected-light petrography and electron microprobe work. Thanks also to the Questa Division of Union Molycorp and the staff of Carson National Forest for their cooperation. D. Champion, A. Cox, B. Criss, G.K. Czamanske, S. Grommé, P.W. Lipman, M. McWilliams, and J.R. O'Neil provided helpful reviews of the manuscript. This paper constitutes a portion of J.T.H.'s Ph.D. thesis at Stanford University.

References

1 R.T. Merrill, Magnetic effects associated with chemical changes in igneous rocks, Geophys. Surv. 2, 277, 1975.
2 I.S.E. Carmichael and J. Nicholls, Iron-titanium oxides and oxygen fugacities in volcanic rocks, J. Geophys. Res. 72, 4665, 1967.
3 E. Larson, Mituko Ozima, Minoru Ozima, T. Nagata and D. Strangway, Stability of remanent magnetization of igneous rocks, Geophys. J.R. Astron. Soc. 17, 263, 1969.
4 C.S. Grommé, T.L. Wright and D.L. Peck, Magnetic properties and oxidation of iron-titanium oxide minerals in Alae and Makaopuhi lava lakes, Hawaii, J. Geophys. Res. 74, 5277, 1969.
5 J. Verhoogen, Oxidation of iron-titanium oxides in igneous rocks, J. Geol. 70, 168, 1962.
6 P.W. Readman and W. O'Reilly, Magnetic properties of oxidised (cation deficient) titanomagnetites, J. Geomagn. Geoelectr. 24, 69, 1972.
7 S. Grommé, E.A. Mankinen, M. Marshall, and R.S. Coe, Geomagnetic paleointensities by the Thelliers' method from submarine pillow basalts: effects of seafloor weathering, J. Geophys. Res. 84, 3553, 1979.
8 K. Kobayashi, An experimental demonstration of the production of chemical remanent magnetization with Cu-Co alloy, J. Geomagn. Geoelectr. 12, 148, 1961.
9 F.D. Stacey and S.K. Banerjee, The Physical Principles of Rock Magnetism, p. 128, Elsevier, Amsterdam, 1974.
10 J.M. Ade-Hall, H.C. Palmer and T.P. Hubbard, The magnetic and opaque petrological response of basalts to regional hydrothermal alteration, Geophys. J.R. Astron Soc. 24, 137, 1971.
11 S.A. Vincenz, K. Yaskawa, and J.M. Ade-Hall, Origin of the magnetization of the Wichita mountains granites, Oklahoma, Geophys. J.R. Astron. Soc. 42, 21, 1975.
12 R.T. Merrill and C.S. Grommé, Nonreproducible self-reversal of magnetization in diorite, J. Geophys. Res. 74, 2014, 1969.
13 A.F. Buddington and D.H. Lindsley, Iron-titanium oxide minerals and synthetic equivalents, J. Petrol. 5, 310, 1964.
14 R.E. Criss and D.E. Champion, Magnetic properties of granitic rocks from the southern half of the Idaho batholith: influences of hydrothermal alteration and implications for aeromagnetic interpretation, J. Geophys. Res. 89, 7061, 1984.
15 H.P. Taylor, Jr., Water/rock interactions and the origin of H_2O in granitic batholiths, J. Geol. Soc. London 133, 509, 1977.
16 H.P. Taylor, Jr. and R.W. Forester, An oxygen and hydrogen isotope study of the Skaergaard intrusion and its country rocks: a description of a 55-m.y.-old fossil hydrothermal system, J. Petrol. 20, 355, 1979.
17 R.E. Criss and H.P. Taylor, Jr., An $^{18}O/^{16}O$ and D/H study of Tertiary hydrothermal systems in the southern half of the Idaho batholith, Geol. Soc. Am. Bull. 94, 640, 1983.
18 Y. Bottinga and M. Javoy, Comments on oxygen istope geothermometry, Earth Planet. Sci. Lett. 20, 250, 1973.
19 Y. Bottinga and M. Javoy, Oxygen isotope partitioning among the minerals in igneous and metamorphic rocks, Rev. Geophys. Space Phys. 13, 401, 1975.
20 M. Javoy, Stable isotopes and geothermometry, J. Geol. Soc. London 133, 609, 1977.
21 J.T. Hagstrum and P.W. Lipman, Paleomagnetism of the structurally deformed Latir volcanic field, northern New Mexico: relations to formation of the Questa caldera and development of the Rio Grande rift, J. Geophys. Res., in press, 1986.
22 J.T. Hagstrum, P.W. Lipman, and D.P. Elston, Paleomagnetic evidence bearing on the structural development of the Latir volcanic field near Questa, New Mexico, J. Geophys. Res. 87, 7833, 1982.
23 J.C. Reed, Jr., Precambrian rocks near the intersection of the Jemez zone and Rio Grande rift, New Mexico, Geol. Soc. Am. Abstr. Prog. 15, 321, 1983.
24 P.W. Lipman, The Miocene Questa caldera, northern New Mexico: relation to batholith emplacement and associated molybdenum mineralization, in: The Genesis of Rocky Mountain Ore Deposits: Changes with Time and Tectonics, Proceedings of the Denver Region Exploration Geologists Society Symposium, p. 133, 1983.
25 R.W. Leonardson, G. Dunlop, V.L. Starquist, G.P. Bratton, J.W. Meyer, L.W. Osborne, S.A. Atkin, P.A. Molling, R.F. Moore and S.D. Olmore, Preliminary geology and molybdenum deposits at Questa, New Mexico, in: The Genesis of Rocky Mountain Ore Deposits: Changes with Time and Tectonics, Proceedings of the Denver Region Exploration Geologists Society Symposium, p. 151, 1983.
26 K. Shidata and S. Ishihara, K-Ar dates on biotite from Questa mine, New Mexico, USA, Bull. Geol. Surv. Jpn. 19, 35, 1968.
27 A.W. Laughlin, W.A. Rehrig and R.L. Mauger, K-Ar chronology and sulphur and strontium isotope ratios at the Questa mine, New Mexico, Econ. Geol. 64, 903, 1969.
28 P.W. Lipman, Volcano-tectonic setting of Tertiary ore deposits, southern Rocky Mountains, in: Relations of Tectonics to Ore Deposits in the Southern Cordillera, W.R. Dickinson and W.D. Payne, eds., Ariz. Geol. Soc. Digest 14, 199, 1981.
29 P.W. Lipman and H.H. Mehnert, Late Cenozoic basaltic volcanism and development of the Rio Grande depression in the southern Rocky Mountains, in: Cenozoic History of the Southern Rocky Mountains, B.F. Curtis, ed., Geol. Soc. Am. Mem. 144, 119, 1975.
30 J.L. Kirshvink, The least-squares line and plane and analysis of palaeomagnetic data, Geophys. J.R. Astron. Soc. 62, 699, 1980.
31 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A 217, 295, 1953.
32 T.C. Onstott, Application of the Bingham distribution function in paleomagnetic studies, J. Geophys. Res. 85, 1500, 1980.
33 J. Borthwick and R.S. Harmon, A note regarding ClF_3 as an alternative to BrF_5 for oxygen isotope analysis, Geochim. Cosmochim. Acta 46, 1665, 1982.
34 J. Bigeleisen, M.L. Perlman, and H.C. Prosser, Conversion of hydrogenic materials to hydrogen for isotopic analysis, Anal. Chem. 24, 1356, 1952.
35 H. Craig, Isotopic variations in meteoric waters, Science 133, 1702, 1961.
36 R.T. Gregory and H.P. Taylor, Jr., An oxygen isotope

profile in a section of Cretaceous oceanic crust, J. Geophys. Res. 86, 2737, 1981.
37 D. Norton and H.P. Taylor, Jr., Quantitative simulation of the hydrothermal systems of crystallizing magmas on the basis of transport theory and oxygen isotope data: an analysis of the Skaergaard intrusion, J. Petrol. 20, 421, 1979.
38 J.R. O'Neil and R.N. Clayton, Oxygen isotope geothermometry, in: Isotopic and Cosmic Chemistry, H. Craig, S.L. Miller and G.J. Wasserburg, eds., p. 157, North-Holland, Amsterdam, 1964.
39 J.R. O'Neil and H.P. Taylor, Jr., The oxygen isotope and cation chemistry of feldspars, Am. Mineral. 52, 1414, 1967.
40 R.N. Clayton, J.R. O'Neil and T.K. Mayeda, Oxygen isotope exchange between quartz and water, J. Geophys. Res. 77, 3057, 1972.
41 R.H. Brigham, K-feldspar genesis and stable isotope relations of the Papoose Flat pluton, Inyo Mountains, California, Ph.D. Thesis, Stanford University, 1984.
42 M. Javoy, S. Fourcade and C.J. Allegre, Graphical method for examinations of $^{18}O/^{16}O$ fractionations in silicate rocks, Earth Planet. Sci. Lett. 10, 12, 1970.
43 Y.T. Wu, M. Fuller and V.A. Schmidt, Microanalysis of N.R.M. in a granodiorite intrusion, Earth Planet. Sci. Lett. 23, 275, 1974.
44 M.E. Evans, M.W. McElhinny and A.C. Gifford, Single-domain magnetite and high coercivities in a gabbroic intrusion, Earth Planet. Sci. Lett. 4, 142, 1968.
45 R.B. Hargraves and W.M. Young, Source of stable remanent magnetism in Lambertville diabase, Am. J. Sci. 267, 1161, 1969.
46 G.S. Murthy, M.E. Evans, and D.I. Gough, Evidence of single-domain magnetite in the Michikamau anorthosite, Can. J. Earth Sci. 8, 361, 1971.
47 R. Dodson, J.R. Dunn, M. Fuller, I. Williams, H. Ito, V.A. Schmidt and Y. Wu, Paleomagnetic record of a late Tertiary field reversal, Geophys. J.R. Astron. Soc. 53, 373, 1978.
48 J.R. O'Neil and H.P. Taylor, Jr., The oxygen isotope and cation exchange chemistry of feldspars, Am. Mineral. 52, 1414, 1967.
49 J.F. Diehl, M.E. Beck, Jr., S. Beske-Diehl, D. Jacobson and B.C. Hearn, Jr., Palaeomagnetism of the Late Cretaceous–Early Tertiary north-central Montana alkalic province, J. Geophys. Res. 88, 10593, 1983.